Therefore, $\frac{ds}{dt} = 20t + 50$

To find the distance function s one has to integrate the derivative function.

That is, $s = \int (20t + 50)\, dt$

$$s = 10t^2 + 50t + c$$

The distance covered by the train is zero when time is zero. Let us use this initial condition $s = 0$ at $t = 0$ to determine the value c of the constant of integration.

$$\Rightarrow s = 10t^2 + 50t + c \quad \Rightarrow c = 0$$

Therefore, $s = 10t^2 + 50t$

The distance covered by the train in 2 hours (5pm-3pm) is given by substituting $t = 2$ in the above equation, we get

$$s = 10(2)^2 + 50(2) = 140 \text{ km.}$$

Example 11.12

The rate of change of weight of person w in kg with respect to their height h in centimetres is given approximately by $\frac{dw}{dh} = 4.364 \times 10^{-5} h^2$. Find weight as a function of height. Also find the weight of a person whose height is 150 cm.

Solution

The rate of change of weight with respect to height is

$$\frac{dw}{dh} = 4.364 \times 10^{-5} h^2$$

$$w = \int 4.364 \times 10^{-5} h^2 \, dh$$

$$w = 4.364 \times 10^{-5} \left(\frac{h^3}{3} \right) + c$$

One can obviously understand that the weight of a person is zero when height is zero.

Let us find the value c of the constant of integration by substituting the initial condition $w = 0$, at $h = 0$, in the above equation

$$w = 4.364 \times 10^{-5} \left(\frac{h^3}{3} \right) + c \Rightarrow c = 0$$

The required relation between weight and height of a person is

$$w = 4.364 \times 10^{-5} \left(\frac{h^3}{3} \right)$$

When the height h =150cm,

$$w = 4.364 \times 10^{-5} \left(\frac{150^3}{3} \right)$$

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